

LINUX LAB MACHINE

Swapping Kali Linux for Ubuntu

Building a Dedicated Linux Environment for IT Infrastructure Learning

Project Title	Linux Lab Machine: Swapping Kali Linux for Ubuntu
Date	August 27–28, 2026
Analyst	Bryan Sykes
Hardware	ASUS X550ZA — AMD A8-7100, 8GB DDR3, 256GB Fanxiang S101Q SATA SSD
Previous OS	Kali Linux
New OS	Ubuntu Desktop 26.04 LTS (Resolute Raccoon)
Hostname	sykes-ubuntu
SSH User	mrbrsnykes
Remote Access	SSH from primary Windows desktop workstation
Coverage	OS Migration, Hardware Background, SSH Remote Access, Troubleshooting Log

Analyst: Bryan Sykes | Home Lab Portfolio Project

1. Project Overview

This project documents the migration of a dedicated physical lab machine. An ASUS X550ZA laptop from Kali Linux to Ubuntu Desktop 26.04 LTS (Resolute Raccoon). The migration was a deliberate decision driven by career focus: Kali Linux is a specialized penetration testing distribution, not a general-purpose operating system, and is not the appropriate platform for building foundational IT infrastructure and data center skills.

The primary objective is to establish a dedicated physical Linux machine that can be administered remotely over SSH from a primary workstation, mirroring real-world server management practices where direct console access is rarely used. This machine runs alongside SYKESHOMESERVER in a personal home lab ecosystem, providing a hands-on Linux learning environment that complements the virtual machine work documented in the Active Directory Home Lab series.

Ubuntu was selected for its dominance in enterprise and federal IT environments. APT package management, systemctl service management, and the broader Ubuntu ecosystem are directly transferable skills for help desk, SOC Analyst Tier 1, and data center technician roles the federal contracting positions this portfolio targets. The planned learning progression on this machine is: Ubuntu Desktop → Ubuntu Server → Rocky Linux (RHEL equivalent).

This project is distinct from spinning up a virtual machine instance of Linux. The deliberate choice to use a dedicated physical machine reflects the reality of enterprise environments, where Linux runs on bare metal servers. The hardware-level challenges encountered during this project were boot configuration, driver compatibility, network interface naming, and privilege escalation. Some of the more representative of real IT support scenarios than a clean VM installation.

2. Hardware Background

The ASUS X550ZA is a 2015-era consumer laptop repurposed as a dedicated lab machine. It was previously used for music production and later repurposed through a hardware upgrade project documented separately in this portfolio. The original 1TB HDD was replaced with a 256GB SATA SSD, and an initial Kali Linux installation was performed. This migration builds on that baseline.

2.1 Hardware Specifications

Component	Specification
Model	ASUS X550ZA (manufactured June 2015)
CPU	AMD A8-7100 (4-core, ~1.8GHz base, Kaveri mobile APU)
RAM	8GB DDR3
Storage	256GB Fanxiang S101Q SATA SSD (upgraded from original 1TB HDD)
Display	15.6" 1366×768
Wired Network	Realtek Gigabit Ethernet (interface: enp4s0)
Wireless Network	MediaTek MT7630E 802.11b/g/n drivers installed via apt upgrade post-install
Known Issue	Trackpad non-functional (External USB mouse required for direct physical interaction)
Previous OS	Kali Linux (installed during Phase 1 hardware repurposing project)

3. Operating System Selection

3.1 Why Ubuntu Over Kali Linux

Kali Linux is purpose-built for penetration testing and offensive security. While valuable as a security tool, it is not designed as a general-purpose daily driver and is an anti-pattern outside of dedicated security testing environments. For building foundational IT infrastructure skills, a general-purpose Linux distribution is the correct foundation.

Criteria	Kali Linux	Ubuntu 26.04 LTS
Primary Purpose	Penetration testing	General-purpose OS
Enterprise Prevalence	Security labs only	Dominant in federal/enterprise IT
Appropriate as Daily Driver	No	Yes
Career Relevance (IT Infra)	Indirect	Direct
Community / Documentation	Security-focused	Largest Linux community globally

3.2 Learning Progression Plan

Phase	Platform	Focus Area
Phase 1 (Current)	Ubuntu Desktop 26.04 LTS	Linux fundamentals, CLI, APT, systemctl, SSH, file permissions
Phase 2	Ubuntu Server	Headless administration, services, cron, UFW, LVM, log management
Phase 3	Rocky Linux (RHEL)	DNF/YUM package management, enterprise Linux conventions for federal environments

4. Installation Process

4.1 ISO Acquisition and Flash Media Preparation

Ubuntu Desktop 26.04 LTS ISO was downloaded from ubuntu.com/download/desktop. Initial flash media preparation was attempted using Balena Etcher, which failed to produce a bootable UEFI USB drive on the X550ZA platform. The final successful flash was performed using Rufus 4.15 with the configuration shown below. The full Etcher failure and USB recovery process is documented in Section 6.

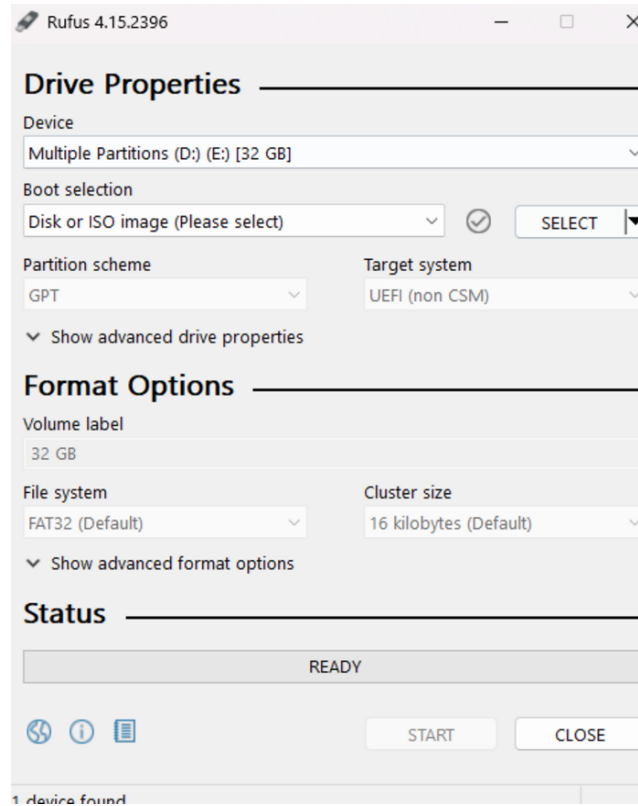


Figure 1 — Rufus 4.15 configuration used for successful Ubuntu flash (GPT / UEFI non-CSM)

Rufus Setting	Value
Device	32GB USB Flash Drive
Boot Selection	Ubuntu Desktop 26.04 LTS ISO
Partition Scheme	GPT
Target System	UEFI (non-CSM)
File System	FAT32 (Default)
Write Mode	Write in ISO Image mode (Recommended)

4.2 BIOS Configuration

The following BIOS settings were verified on the ASUS X550ZA before booting from the USB drive. BIOS is accessed by pressing F2 immediately on power-on. The boot menu for one-time device selection is accessed via ESC.

- Fast Boot: Disabled — bypasses USB device detection during POST
- Secure Boot: Disabled — blocks unsigned Linux bootloaders if enabled
- Boot Mode: UEFI — required to match the GPT/UEFI flash created by Rufus
- Boot Order: UEFI: Flash Disk set as position 1, above the internal SSD

4.3 Ubuntu Installer Walkthrough

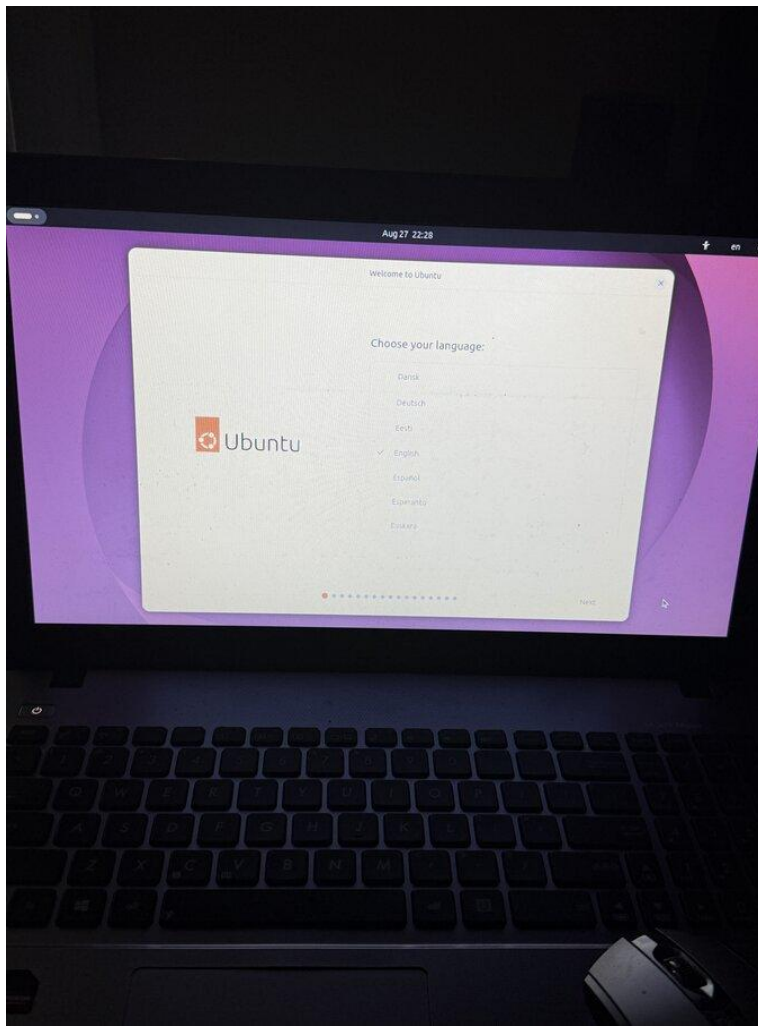


Figure 2 — Ubuntu 26.04 installer language selection screen on the ASUS X550ZA

Installer Screen	Selection / Notes
Language	English
Keyboard Layout	English (US)
Network	Not connected — Wi-Fi drivers absent; installation completed offline
Applications	Default selection (essentials, web browser, basic utilities)
Third-Party Software	Unchecked — drivers installed post-install via apt upgrade
Installation Type	Erase disk and install Ubuntu — fully wiped Kali Linux from SSD
Encryption	No encryption — lab machine, adds unnecessary complexity
Timezone	America/New_York (Eastern)
Username	mrbyansykes
Hostname	sykes-ubuntu
Login	Require password to log in — enabled

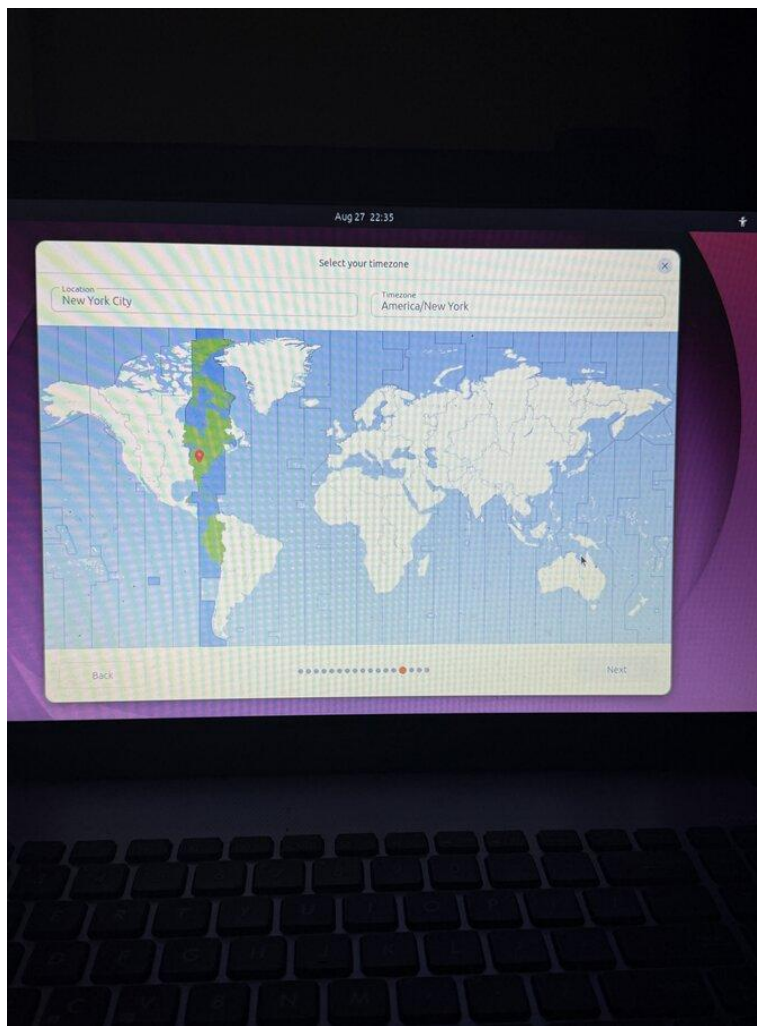


Figure 3 — Timezone selection screen showing America/New_York — correct for Alexandria, VA

5. Post-Installation Configuration

5.1 Sudo Privileges & Recovery Mode Fix

After first boot, the user account `mrbyansykes` did not have `sudo` privileges, preventing execution of administrative commands. Ubuntu locks the root account by default. `su root` and `su -c` both returned authentication failures. The fix was performed via Recovery Mode:

- Rebooted and accessed GRUB menu → Advanced options for Ubuntu → Recovery mode
- Selected "root — Drop to root shell prompt" from the Recovery Menu
- Added user to the `sudo` group:

```
usermod -aG sudo mrbyansykes
```
- Reset the user password:

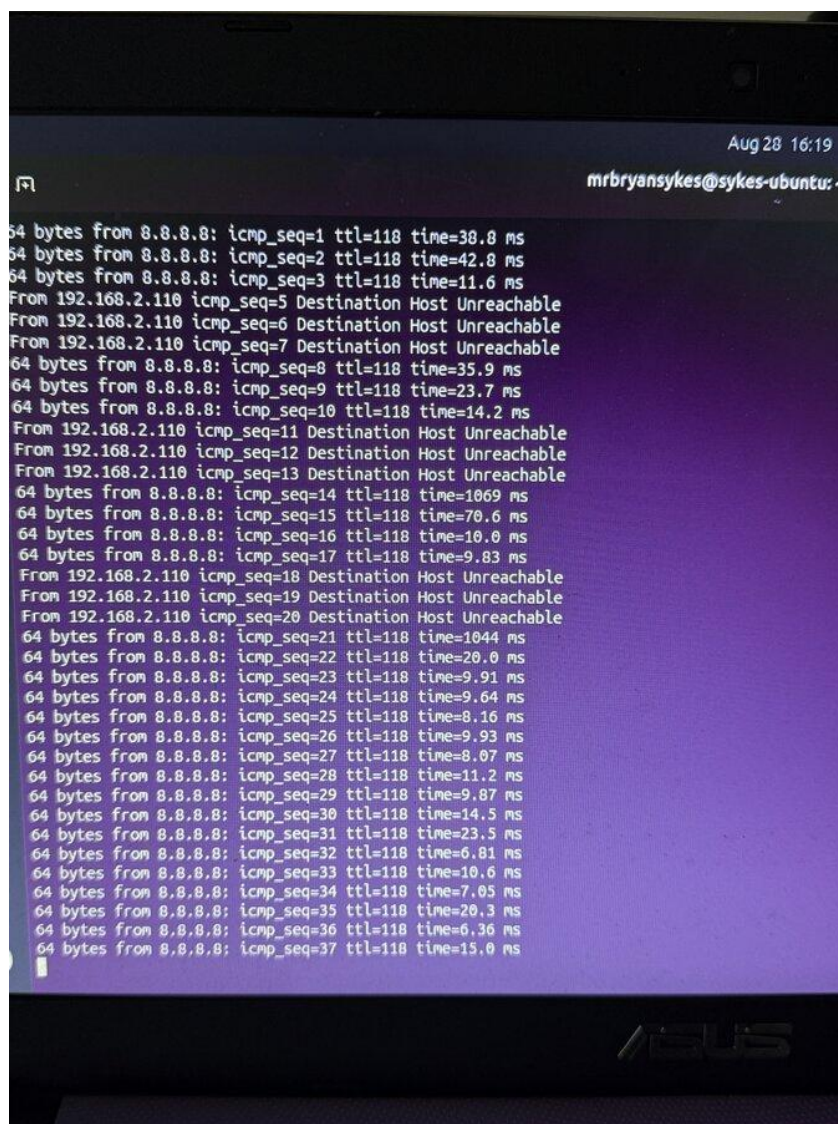
```
passwd mrbyansykes
```
- Exited and resumed normal boot. Verified with:

```
sudo whoami
```

Expected and received output: `root` — confirming `sudo` access was restored.

5.2 Network Connectivity: Wired Ethernet

Wi-Fi drivers were unavailable at first boot. A wired ethernet cable was connected to a Meshforce mesh node. The initial connection showed a 169.254.x.x self-assigned APIPA address, indicating the cable was plugged into the WAN/uplink port rather than a LAN port. After switching to the correct LAN port, a valid 192.168.x.x DHCP lease was obtained and internet connectivity was confirmed via ping.



```
Aug 28 16:19
mrbryansykes@sykes-ubuntu: ~

64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=38.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=42.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=11.6 ms
From 192.168.2.110 icmp_seq=5 Destination Host Unreachable
From 192.168.2.110 icmp_seq=6 Destination Host Unreachable
From 192.168.2.110 icmp_seq=7 Destination Host Unreachable
64 bytes from 8.8.8.8: icmp_seq=8 ttl=118 time=35.9 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=118 time=23.7 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=118 time=14.2 ms
From 192.168.2.110 icmp_seq=11 Destination Host Unreachable
From 192.168.2.110 icmp_seq=12 Destination Host Unreachable
From 192.168.2.110 icmp_seq=13 Destination Host Unreachable
64 bytes from 8.8.8.8: icmp_seq=14 ttl=118 time=1069 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=118 time=70.6 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=118 time=10.0 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=118 time=9.83 ms
From 192.168.2.110 icmp_seq=18 Destination Host Unreachable
From 192.168.2.110 icmp_seq=19 Destination Host Unreachable
From 192.168.2.110 icmp_seq=20 Destination Host Unreachable
64 bytes from 8.8.8.8: icmp_seq=21 ttl=118 time=1044 ms
64 bytes from 8.8.8.8: icmp_seq=22 ttl=118 time=20.0 ms
64 bytes from 8.8.8.8: icmp_seq=23 ttl=118 time=9.91 ms
64 bytes from 8.8.8.8: icmp_seq=24 ttl=118 time=9.64 ms
64 bytes from 8.8.8.8: icmp_seq=25 ttl=118 time=8.16 ms
64 bytes from 8.8.8.8: icmp_seq=26 ttl=118 time=9.93 ms
64 bytes from 8.8.8.8: icmp_seq=27 ttl=118 time=8.07 ms
64 bytes from 8.8.8.8: icmp_seq=28 ttl=118 time=11.2 ms
64 bytes from 8.8.8.8: icmp_seq=29 ttl=118 time=9.87 ms
64 bytes from 8.8.8.8: icmp_seq=30 ttl=118 time=14.5 ms
64 bytes from 8.8.8.8: icmp_seq=31 ttl=118 time=23.5 ms
64 bytes from 8.8.8.8: icmp_seq=32 ttl=118 time=6.81 ms
64 bytes from 8.8.8.8: icmp_seq=33 ttl=118 time=10.6 ms
64 bytes from 8.8.8.8: icmp_seq=34 ttl=118 time=7.05 ms
64 bytes from 8.8.8.8: icmp_seq=35 ttl=118 time=20.3 ms
64 bytes from 8.8.8.8: icmp_seq=36 ttl=118 time=6.36 ms
64 bytes from 8.8.8.8: icmp_seq=37 ttl=118 time=15.0 ms
```

Figure 4 — Consistent ping responses to 8.8.8.8 confirming network connectivity over wired ethernet

5.3 System Update

With sudo restored and ethernet connectivity confirmed, a full system update was performed:

```
sudo apt update && sudo apt upgrade -y
```

The upgrade installed 317 packages including 179 LTS security updates (1,209MB download, 379MB disk space). Critically, the MediaTek MT7630E wireless drivers were installed automatically as part of this upgrade, enabling Wi-Fi without manual driver installation. After the upgrade, the machine was switched to Wi-Fi and the ethernet cable removed.

5.4 OpenSSH Server Installation

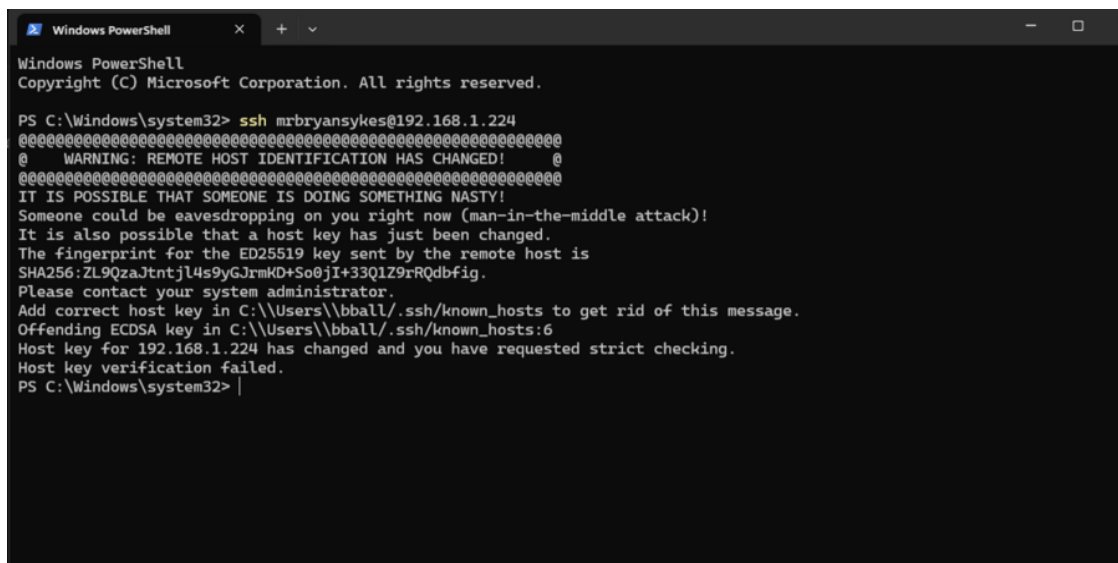
SSH server was installed and enabled for persistent remote access:

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh      # confirmed: active (running)
ip a                          # obtained local IP: 192.168.1.224
```

5.5 SSH Remote Access Test

SSH access was tested from the primary Windows desktop workstation using PowerShell. An initial attempt returned a host key mismatch warning because the IP 192.168.1.224 had previously been associated with the Kali Linux installation. The stale entry was cleared:

```
ssh-keygen -R 192.168.1.224
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> ssh mrbryansykes@192.168.1.224
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@   WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!   @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:ZL9QzaJtntjL4s9yGJrmKD+So0jI+33Q1Z9rRQdbfig.
Please contact your system administrator.
Add correct host key in C:\Users\bball\.ssh\known_hosts to get rid of this message.
Offending ECDSA key in C:\Users\bball\.ssh\known_hosts:6
Host key for 192.168.1.224 has changed and you have requested strict checking.
Host key verification failed.
PS C:\Windows\system32> |
```

Figure 5 — SSH host key mismatch warning on first connection attempt — resolved by clearing known hosts entry

After clearing the known hosts entry, the connection succeeded:

```
ssh mrbryansykes@192.168.1.224
```

```

mrbrvansykes@sykes-ubuntu X + v
* Support: https://ubuntu.com/pro

3 devices have a firmware upgrade available.
Run 'fwupdmgtr get-upgrades' for more information.

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

3 devices have a firmware upgrade available.
Run 'fwupdmgtr get-upgrades' for more information.

mrbrvansykes@sykes-ubuntu:~$ |

```

Figure 6 — Successful SSH session from Windows desktop — PowerShell title bar shows mrbrvansykes@sykes-ubuntu

The machine is now fully operable without direct physical interaction. All administration is performed remotely over SSH from the primary workstation.

6. Troubleshooting Log

This section documents every issue encountered during the installation process with root cause analysis and resolution. Troubleshooting real-world problems is a core IT competency, these are not failures, they are demonstrations of diagnostic methodology applied to genuine hardware and software challenges.

Issue 1 — System Hung on ASUS Logo When Booting from USB

Field	Detail
Symptom	Machine powered on, ASUS logo appeared, and the system hung indefinitely. Eventually fell back to booting Kali Linux from the internal SSD.
Suspected Cause	Fast Boot or Secure Boot enabled in BIOS, blocking USB device detection
Finding	Fast Boot and Secure Boot were both already disabled. Boot order was correct. Root cause: Balena Etcher failed to write a valid UEFI bootloader for the X550ZA platform.
Resolution	Switched from Balena Etcher to Rufus 4.15 with GPT partition scheme and UEFI (non-CSM) target system. Machine booted into Ubuntu installer on the first attempt.

Issue 2 — USB Drive Became Write-Protected After Etcher Flash

Field	Detail
Symptom	After the failed Etcher flash, the USB drive could not be reformatted. Windows reported the drive as write-protected. No physical write-protect switch was present.
Root Cause	Etcher writes Linux partition layouts (including ext4 partitions) that Windows cannot recognize. Windows interprets unrecognized layouts as write-protected.

Resolution	Used Windows Diskpart to clear the write-protection flag and rebuild the partition table. Full procedure documented below.
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Diskpart USB Recovery Procedure:

```
diskpart
list disk
select disk #           (identify USB by size - verify carefully before proceeding)
attributes disk clear readonly
clean                   (WARNING: immediately destroys all data on selected disk)
convert mbr
create partition primary
active
format fs=fat32 quick
assign
exit
```

Issue 3 — "No Volume Detected" Error During Diskpart Format

Field	Detail
Symptom	After attributes disk clear readonly, running format fs=fat32 quick returned "No volume detected."
Root Cause	Disk had no valid partition table. Windows requires an active partition before it can execute a format command.
Resolution	Run clean first to wipe the partition table, then create partition primary to build a fresh one before formatting.

Issue 4 — "Selected Disk Is Not a Fixed MBR Disk" Error

Field	Detail
Symptom	The active command in Diskpart returned: "The selected disk is not a fixed MBR disk. The ACTIVE command can only be used on fixed MBR disks."
Root Cause	Etcher had formatted the USB using GPT. The active command only applies to MBR-formatted disks — GPT uses a different boot configuration model.
Resolution	Run convert mbr after clean and before create partition primary. This converts the scheme from GPT to MBR, allowing active to execute successfully.

Issue 5 — Windows Format Prompts and "D:\ Is Not Accessible" After Rufus Flash

Field	Detail
Symptom	After Rufus completed, Windows displayed "Location is not available — D:\ is not accessible" and prompted to format the drive.
Root Cause	Normal Windows behavior. Ubuntu installer creates partitions using Linux filesystems (squashfs, etc.) that Windows cannot read.
Resolution	Dismissed all popups without formatting. The Rufus Status bar showed READY, confirming the flash was successful. Proceeded directly to booting from USB on the ASUS.

Issue 6 — Wi-Fi Not Detected During Installation

Field	Detail
Symptom	The Ubuntu installer showed no available Wi-Fi networks. The Wi-Fi option was visible but could not scan or connect.
Root Cause	The MediaTek MT7630E wireless driver is not included in the Ubuntu 26.04 live installer environment. A network connection is required to download it — creating a chicken-and-egg situation.
Resolution	Completed installation offline. Essential packages install from USB. Wireless drivers were installed automatically during the first apt upgrade over wired ethernet post-install.

Issue 7 — Sudo Not Available After First Boot

Field	Detail
Symptom	Running sudo returned "command not found." su root and su -c both returned authentication failures.
Root Cause	User account was created without administrator privileges. The root account is locked by default in Ubuntu 26.04 — su root is non-functional without first unlocking it.
Resolution	Used Ubuntu Recovery Mode via GRUB to access a root shell. Ran usermod -aG sudo mrbryansykes and passwd mrbryansykes. Verified with sudo whoami returning root after reboot.

Issue 8 — Ethernet Showing 169.254.x.x Self-Assigned IP (No Internet)

Field	Detail
Symptom	Interface enp4s0 showed IP 169.254.70.10 (APIPA self-assigned). Ping to 8.8.8.8 returned "Network is unreachable."
Root Cause	Ethernet cable was plugged into the WAN/uplink port on the Meshforce mesh node rather than a LAN port. WAN ports do not serve DHCP to client devices.
Resolution	Reconnected to a LAN port on the mesh node. A valid 192.168.x.x DHCP lease was obtained. Connectivity confirmed via sustained ping to 8.8.8.8.

Issue 9 — SSH Host Key Mismatch Warning

Field	Detail
Symptom	First SSH attempt from the Windows desktop returned "WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED" and refused the connection with "Host key verification failed."
Root Cause	The workstation known_hosts file contained the old SSH host key for 192.168.1.224 from the previous Kali Linux installation. Ubuntu generates new SSH host keys on installation.
Resolution	Cleared the stale entry: ssh-keygen -R 192.168.1.224. Subsequent SSH connection accepted the new Ubuntu host key and succeeded.

7. Lab Ecosystem

sykes-ubuntu operates as part of a broader home lab environment. The table below shows the current state of all machines:

Machine	Role	OS / Platform
SYKES-DESKTOP	Primary workstation all lab operations originate here and is the primary ssh device	Windows 11 Home
SYKESHOMESERVER	Home lab server running Docker, VMs, Pi-hole, Jellyfin, Pelican Panel	Windows 11 Pro
sykes-ubuntu	Dedicated Linux daily driver and learning environment	Ubuntu Desktop 26.04 LTS
HP Victus	Secondary workstation / portable	Windows 11 Home

The relationship between sykes-ubuntu and SYKESHOMESERVER is intentional — both run Ubuntu, both are administered over SSH from the primary workstation, and both use the same package manager and service management tools. Skills developed on this desktop machine transfer directly to the server environment already running in the lab.

8. Key Lessons and Technical Notes

- Balena Etcher vs. Rufus: Etcher is simple but lacks control over partition scheme and target system mode. For hardware with specific UEFI boot requirements, Rufus with explicit GPT / UEFI (non-CSM) settings is the more reliable choice for creating Linux bootable media on Windows.
- Linux partition layouts are invisible to Windows: Any USB drive flashed with a Linux ISO will appear inaccessible in Windows Explorer. This is expected — Windows cannot read ext4 or squashfs. The Diskpart utility bypasses this by operating at the disk level rather than the filesystem level.
- GPT vs. MBR in USB recovery: Etcher writes GPT partition tables. The Diskpart active command and some legacy format operations require MBR. Running `convert mbr` is the required intermediate step when recovering a Linux-flashed USB on Windows.
- Ubuntu locks the root account by default: This is a security feature. The correct Ubuntu administrative model uses a regular user with sudo privileges — not direct root login. Recovery Mode provides a legitimate path to restore sudo group membership without the root password.
- APIPA addresses (169.254.x.x) indicate DHCP failure: A self-assigned IP in the 169.254.0.0/16 range means the interface could not obtain a DHCP lease. On mesh network nodes, this commonly occurs when connecting to the WAN/uplink port instead of a LAN port.
- SSH host keys are machine-specific: Installing a new OS generates new SSH host keys. Any client that previously connected to the same IP will see a mismatch warning. Clearing the `known_hosts` entry with `ssh-keygen -R <ip>` is the correct resolution in a controlled lab environment.
- Wi-Fi drivers can install automatically via apt upgrade: The MediaTek MT7630E driver was unavailable during the offline installer but was pulled automatically during the first apt upgrade over wired ethernet. The correct workflow: install offline → connect ethernet → upgrade → switch to Wi-Fi.

- A dedicated physical machine is categorically different from a VM: Virtual machines do not expose hardware-level challenges like driver compatibility, boot configuration, BIOS settings, and network interface naming. Working through these challenges on physical hardware is a more direct preparation for data center and infrastructure roles.

9. Next Steps

Action	Purpose
Set DHCP reservation in Meshforce router for sykes-ubuntu MAC address	Ensures machine always receives IP 192.168.1.224 — critical for reliable SSH access without reconnection
Configure UFW firewall	Harden the machine. allow only necessary ports (SSH: 22)
Set up SSH key-based authentication	Disable password SSH login; use key pairs for more secure remote access. Portfolio talking point for security hardening
Learn core Linux CLI skills	apt, systemctl, chmod/chown, grep/awk/sed, cron, df/top/htop, ip, ufw — all directly applicable to help desk and SOC roles
Transition to Ubuntu Server on bare metal	Phase 2 of Linux learning progression — headless server administration mirrors production data center environments
Begin Rocky Linux familiarization	Phase 3 — RHEL-equivalent skills for federal and enterprise environments; DNF/YUM package management

10. Conclusion

This project successfully migrated a physical ASUS X550ZA laptop from Kali Linux to Ubuntu Desktop 26.04 LTS, establishing a dedicated Linux learning environment for IT infrastructure skill development. The migration involved nine documented real-world challenges. Ranging from USB boot failures, write-protected drives, GPT/MBR partition conflicts, sudo privilege issues, DHCP troubleshooting, and SSH host key conflicts. All diagnosed and resolved systematically.

The end state is a functional Ubuntu machine (sykes-ubuntu) that boots independently, connects to the home network over Wi-Fi, and is fully administrable via SSH from the primary Windows desktop workstation without any direct physical interaction. This mirrors the remote administration model used for servers in enterprise and federal data center environments.

The deliberate choice to use a physical dedicated machine rather than a virtual machine reflects a commitment to building skills that transfer directly to real infrastructure work. The troubleshooting challenges encountered and resolved during this project are more representative of real IT support scenarios than a clean virtual machine installation would ever produce.

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